Nicholas Bratvold

nhbratvold@gmail.com | NicholasBratvold.github.io | +1 415 971 3237
github.com/NicholasBratvold | linkedin.com/in/nicholas-bratvold

Experience

Biohub *Python, PyTorch, Machine Learning, Microfluidics, Optical Design, Rapid Prototyping*

Research Engineer I Jan. 2026 - Present

Associate Research Engineer Aug. 2024 - Dec. 2025

- Drive machine-learning, scientific software, and data-platform development across two automated biological imaging programs.

Remoscope: A Label-free Imaging Cytometer

- Architected a PostgreSQL blood-cell database and custom Python API spanning 1,700 patient samples and more than 3 billion unique cell observations across malaria, sepsis, sickle-cell disease, diabetes, and healthy cohorts, enabling researchers to query, annotate, and train on newly acquired samples within an hour of collection.
- Developed an interpretable multiple-instance learning model for label-free sepsis detection that distinguished sepsis from noninfectious ICU and healthy patients with an AUROC of 0.96 under leave-one-patient-out cross-validation on a 51-participant cohort, using only whole-blood images and patient-level labels with a 10-minute turnaround time.
- Trained a self-supervised representation model on one billion blood cells, enabling whole-blood-cell morphology mapping, analysis of disease-associated population shifts, and accelerated annotation of rare cell types.
- Led the AI work supporting Real Time Imaging Systems' first-place Medical Devices award in the Nebius AI Discovery Awards, receiving $100,000 in compute credits.
- Planned and executed the machining, assembly, and testing of eight Remoscope devices for use in research labs and clinics in the USA, Rwanda, and Uganda.

Fish Sorter: Automated Zebrafish Phenotyping

- Fine-tuned a DINOv3 vision transformer using semi-supervised learning to classify known phenotypes and identify novel phenotypic clusters across 30 zebrafish lines from 0-7 days post-fertilization.
- Designed an interactive annotation interface for exploring, grouping, and validating embedding clusters, enabling six biologists to efficiently create labels and refine phenotypic classes.
- Integrated phenotype classification with a robotic sorting system that automatically images, classifies, and physically places 595 fish into assigned wells in 30 minutes, doubling manual throughput and reducing active handling time.

Zen Maker Lab May. 2021 - Dec. 2021

Engineer Project Designer *Python, C++, Unity, Fusion 360, 3D printing*

- Led a project-based engineering program helping students design and build their ambitious ideas and equip them with the skillset to realize them, such as; a battle-bot design kit, a Tesla coil, RL Minecraft agent, VR video game, and a wildlife camera website.

Projects

EleutherAI Sep. 2023 - Apr. 2024

Generative AI Video Model *Python, JAX, OpenCV*

- Optimized an open-source model that generates video through a two-stage architecture consisting of a variational autoencoder and diffusion transformer, supporting a 500x312 resolution at various video lengths.
- Achieved 500x training speed improvement through data parallelism and data preprocessing.
- Optimized hyperparameters to improve generated video quality and prove model scalability.

University of British Columbia Sep. 2021 - Apr. 2022

Portable Fentanyl Quantification Device *Python, OnShape, PCB Design, Microfluidics*

- Integrated a proven three-stage process of chromatography, voltammetry, and computational analysis to produce precise readings of fentanyl and other drug concentrations. Capable of detecting 100x below the lethal dose of fentanyl.
- Designed and validated a 50μL electrochemical flow cell to ensure accurate results. Prioritized a portable design with a GUI to allow ease of transport and increase investor interest in the product.

Education

University of British Columbia BASc Engineering Physics - Class of 2024

Skills

Languages: Python, C++, SQL, Java, MATLAB, JavaScript, HTML/CSS

Technologies & Tools: OpenCV, PyTorch, JAX, PostgreSQL, ROS, Git, Linux, OpenGL, SolidWorks, 3D Printing